D4.1 Natural selection

D4 Continuity and change
Ecosystems

HL

Understandings:

- D4.1.9—Concept of the gene pool
- D4.1.10—Allele frequencies of geographically isolated populations
- D4.1.11—Changes in allele frequency in the gene pool as a consequence of natural selection between individuals according to differences in their heritable traits
- D4.1.12—Differences between directional, disruptive and stabilizing selection
- D4.1.13—Hardy–Weinberg equation and calculations of allele or genotype frequencies
- D4.1.14—Hardy–Weinberg conditions that must be maintained for a population to be in genetic equilibrium
- D4.1.15—Artificial selection by deliberate choice of traits

Key words:

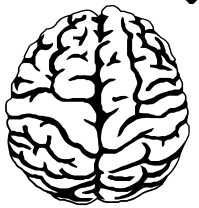
Gene pool
Population bottleneck
Founder effect
Geographic isolation
Heritable
Traits
Neo-Darwinism
Stabilising selection
Disruptive selection
Directional selection
Speciation
Hardy-Weinberg
Genetic drift
Non random mating
Mutation
Gene flow
Natural selection
Artificial selection

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.

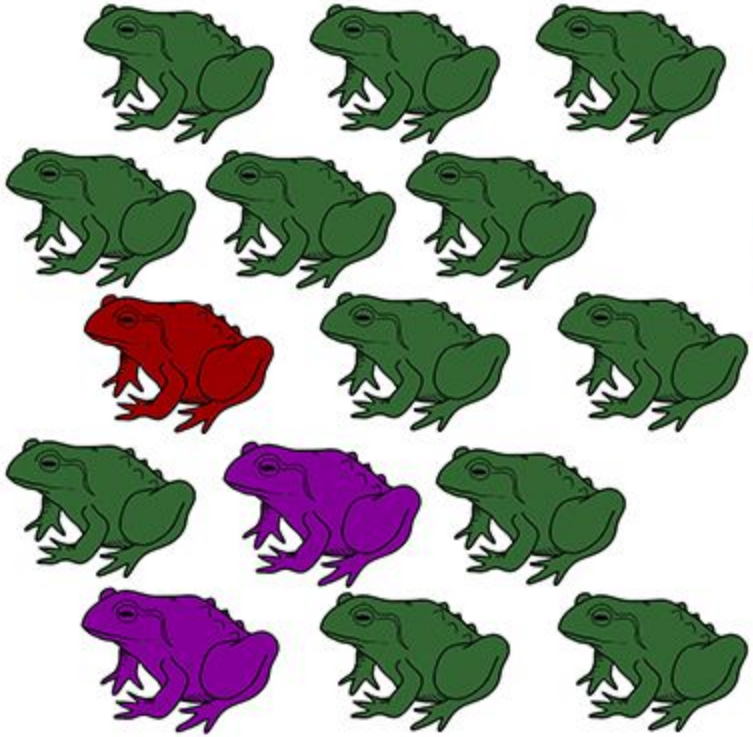


Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.

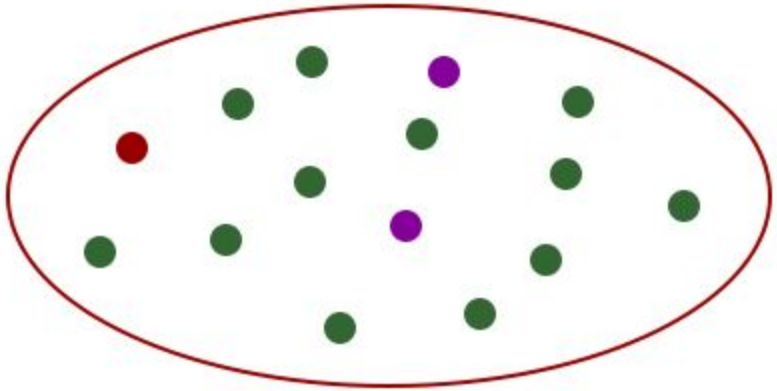


D4.1.9—Concept of the gene pool

a population



gene pool



If populations become reproductively isolated they may form separate **gene pools** (while still remaining of the same species)

D4.1.10—Allele frequencies of geographically isolated populations

If populations of the same species are geographically isolated then the frequency of the alleles in their separate gene pools may be different.

This could be due to **differential selection** but it may be to do with which alleles went into each gene pool.

This can be thought of in terms of the **founder effect** and **population bottlenecks** (more later!)



Genetic drift caused by the population bottlenecks and the founder effect can lead to rare (and problematic) alleles becoming more common.

Databases such as ALFRED can be used to look at allele frequencies.

Table 1. Population-specific prevalence of ten common autosomal recessive disorders.

Disorder	Causative genes	Pathogenic variants[a]	Interethnic variability	Cases per 10,000 individuals							
				Global	EUR	FIN	AJ	LAT	SAS	EAS	AFR
Hyperprolinemia	PRODH, ALDH4A1	207 (7/200)	663	54.9	114.2	23.9	**132.5**	24.1	23.1	0.2	12.7
Hemochromatosis, type 1	HFE	35 (5/30)	335	11.6	**33.5**	12.3	1.4	2.4	0.1	<0.1	1.2
Biotinidase deficiency	BTD	124 (59/65)	309	12.2	18.4	**30.9**	10.8	5	17.7	<0.1	0.9
Stargardt disease	ABCA4	528 (149/379)	113	4.3	4.8	0.2	6.3	3	1.3	2.9	**22.5**
Sickle cell anemia	HBB	62 (44/18)	211	0.6	<0.1	0.1	<0.1	0.1	1.8	0.2	**21.1**
Familial Mediterranean fever	MEFV	95 (15/80)	155	0.1	0.1	<0.1	**15.5**	<0.1	<0.1	0.2	<0.1
Gaucher disease	GBA	93 (34/59)	98	0.1	0.1	0.1	**9.8**	<0.1	<0.1	<0.1	<0.1
Adrenal hyperplasia, congenital, due to 21-hydroxylase deficiency	CYP21A2	49 (10/39)	48	1.1	1.8	0.3	0.2	0.5	0.2	**9.6**	0.3
Infantile hypercalcemia	CYP24A1,SLC34A1	278 (19/259)	18	4.2	8.3	**8.8**	1.1	0.5	2.3	0.8	0.6
Cystic fibrosis	CFTR	408 (183/225)	39	3.1	6	0.6	**7.7**	2.2	1.2	0.2	1.3

Interethnic variability is defined as the fold change between the highest and the lowest population-specific frequency. The highest population-specific prevalence for each disease is indicated in bold.

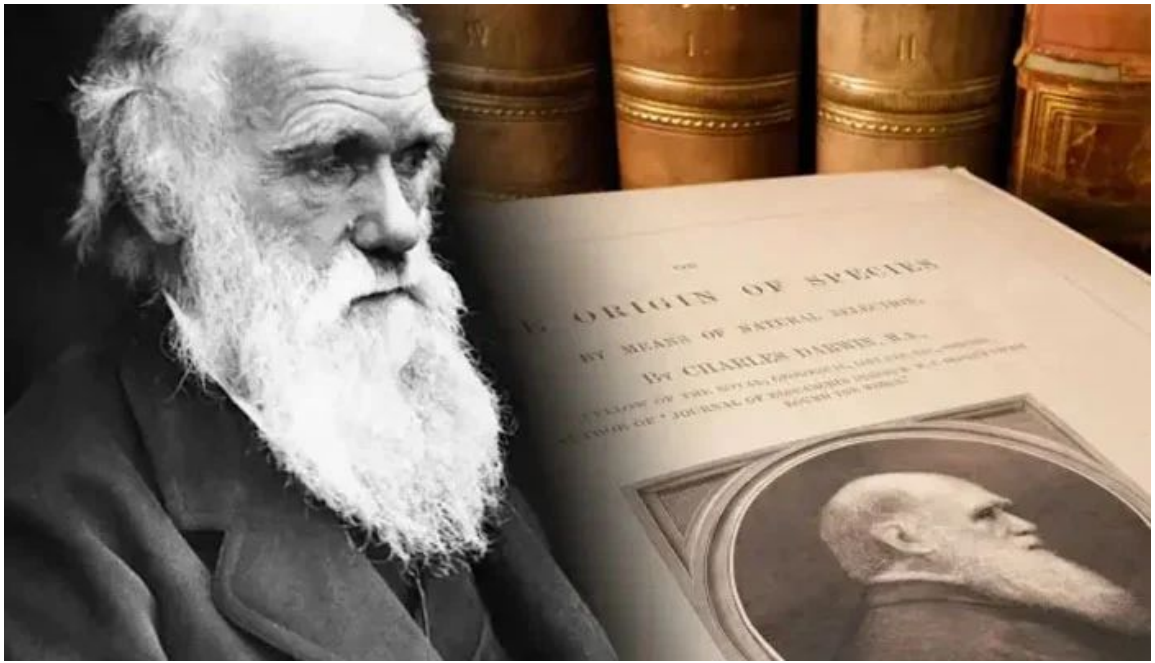
EUR Europeans, *FIN* Finns, *AJ* Ashkenazi Jews, *LAT* Latin Americans, SAS South Asians, *EAS* East Asians, *AFR* Africans.

[a]Values in brackets indicate the number of pathogenic and number of predicted pathogenic variants.

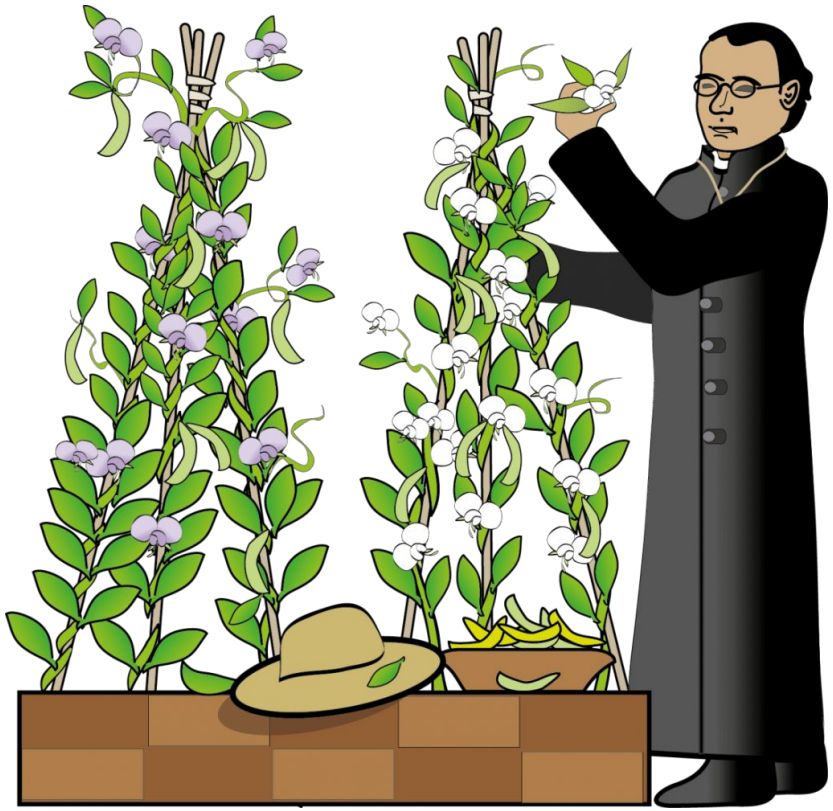
D4.1.11—Changes in allele frequency in the gene pool as a consequence of natural selection between individuals according to differences in their heritable traits

In 1859 Darwin proposed evolution by natural selection

Did not understand about genes, alleles and chromosomes

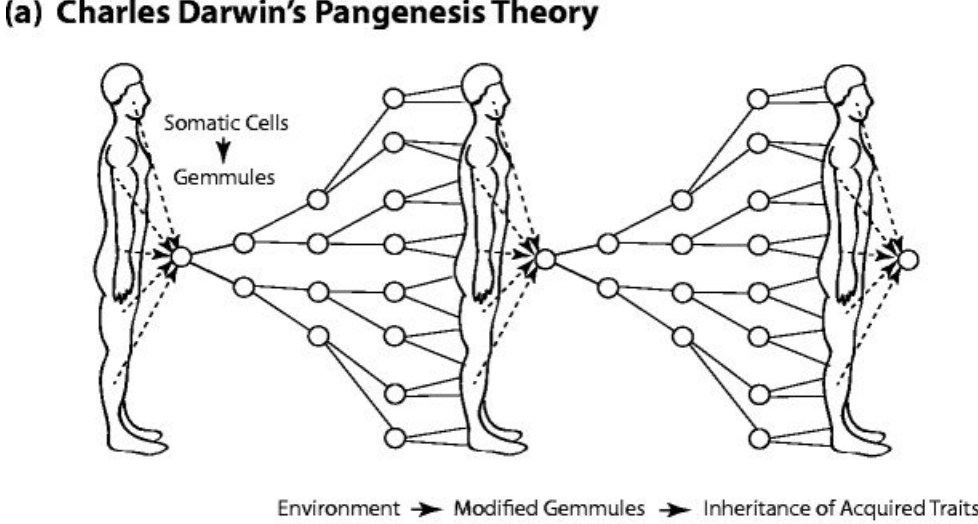


In 1866 Mendel published his findings of the inheritance of traits in pea plants



Germ-plasm theory

In 1893 Weismann proposed that acquired characteristics were expressed in somatic cells whereas only the germ cells carried hereditary information



Neo-Darwinism

Darwin’s theory together with the contributions by Mendel and Weismann is known as **Neo-Darwinism**

D4.1.12—Differences between directional, disruptive and stabilizing selection

There are three patterns of natural selection:

Stabilising selection

Disruptive selection

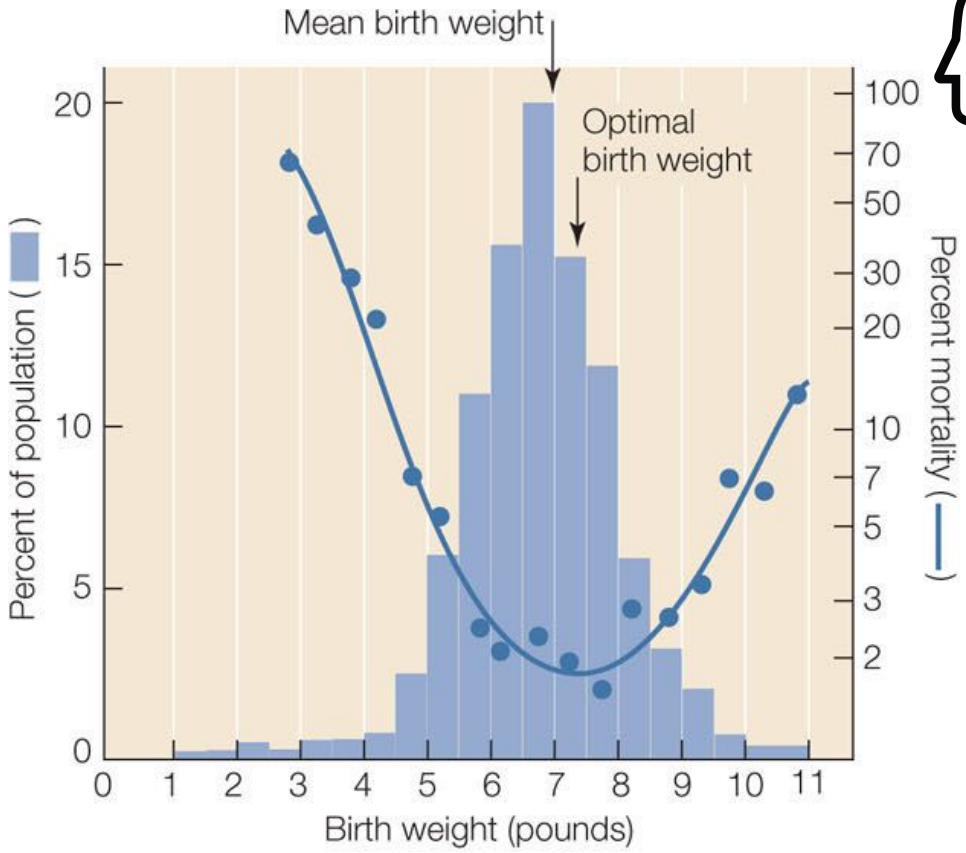
Directional selection

Stabilising selection

The extremes are selected against.

In stable environments the populations tends towards an average.

Features that are common within species and also across species have come from stabilising selection!



What might the reasons be for an average newborn weight of around 3.5kg?



1. Which of the following is NOT stabilizing selection?

A. A population of foxes shifts from mostly red to mostly grey

B. The most common color of rabbit increases after new predators are introduced

C. A population of purple sea urchins stays purple, as starfish eat other colors

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B. The most common color of rabbit increases after new predators are introduced

C. A population of purple sea urchins stays purple, as starfish eat other colors

A is correct. In this case, the foxes are changing in majority from red to grey. This could indicate a number of driving factors, but is *directional selection*, not stabilizing selection. The other two cases represent situations in which the majority was selected for, and increased in frequency, due to the forces of predation.



2. Which of the following was not caused by forces driving stabilizing selection?

A. A species of moth, divided by selection, becomes two species

B. A species of rhino has 2 horns, instead of any other number

C. Most vertebrates with limbs have 4 or 5 toes

2. Which of the following was not caused by forces driving stabilizing selection?

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B. A species of rhino has 2 horns, instead of any other number

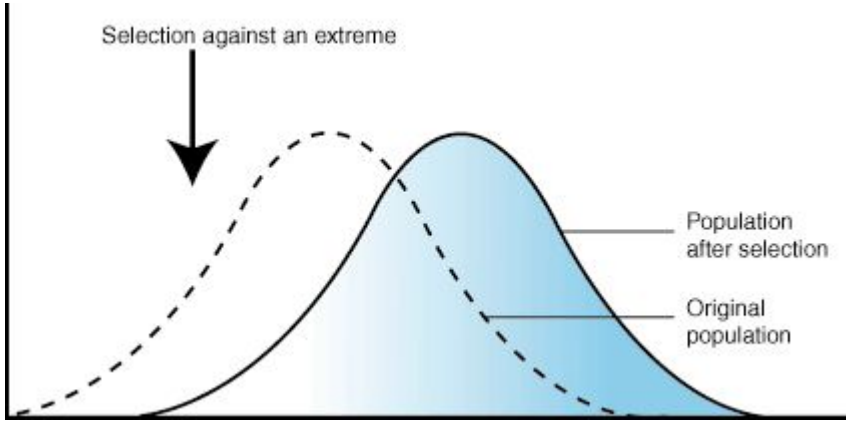
C. Most vertebrates with limbs have 4 or 5 toes

A is correct. Again, anything which selects against the majority is not from stabilizing selection. Here, a group of moths becomes divided, a form of *disruptive selection*. In the other two answers, the number has been determined by stabilizing forces on both sides of the trait. Notice that questions one and two are very similar, but are asked in different ways. Don’t get tricked up by complex wording!

Directional selection

Whole population moves towards an extreme.

Why might giraffes have long necks?





1. Which of the following is an example of directional selection?

A. A population of birds is split into two populations, one with large feet, and one with small.

B. The maximum size of a fish species increases, because a new predator can eat the smaller members.

C. A population of plants evolves to be the same height. Too high and they tip over, too short and they cannot get enough sunlight.

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B is correct. Directional selection exists when one of the extremes of a trait is selected for. In answer A, disruptive selection is at work because population was divided. In answer C, stabilizing selection is at work because plants of a moderate height are selected for. Only in answer B is one side of the spectrum selected for, and the population will continue to get bigger. Eventually, the population may reach a size that is too big to sustain. In this case, size would be under a stabilizing selection, as both high and low values would be selected against.



2. A population of rabbits has coat colors ranging from white to black, with every shade of grey in between. The population is under a strong directional selection. Which of the following could be true of the population after many generations have passed?

A. The same frequency of phenotypes will be seen.

B. An increase of black and dark grey rabbits will take place.

C. Though grey rabbits will perish, both white and black will be successful.

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B is correct. If the black and dark grey rabbits increased, the selection would be in one direction on the spectrum of phenotypic characteristics. This answer could be true, given the details in the question. Answer A cannot be true, if the population is under a form a selection. Without a change in the frequency of phenotypes, it could never be known if a selective pressure exists. Therefore, it is unlikely that A could be true. If C were true, the population would be going through a disruptive selection, not a directional selection.



3. Winglessness is a mutation seen in fruit flies. In nature, these flies are quickly killed. In a captive environment, however, they can function and reproduce with no hardship. A population of fruit flies exists in a captive environment with no predators, and plenty of food. Half the flies have wings, the other half do not. A new predator, the dart frog, is introduced into the captive environment. Which type of selection will the frog put on the phenotype of wings?

A. Stabilizing Selection

B. Directional Selection

C. Disruptive Selection

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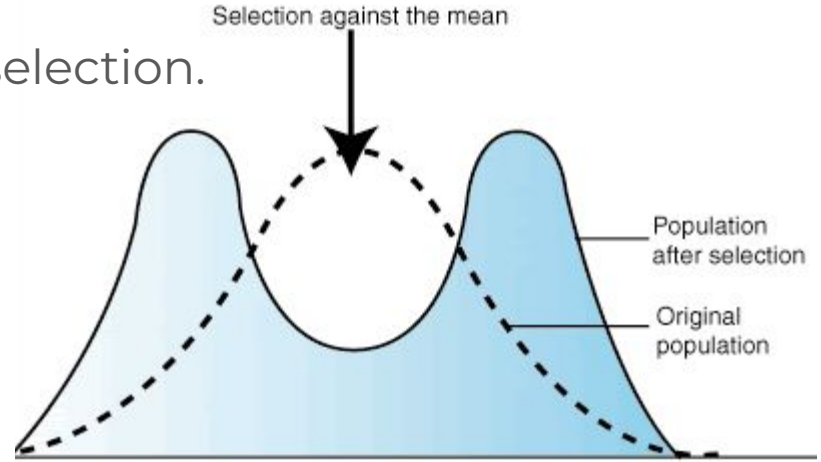
B is correct. The frog, aided by the ease of capturing wingless flies, will quickly eat the wingless flies first. The wingless flies will have less of an ability to reproduce the next generation, and the population will be mostly winged flies after a few generations. This is a form of directional selection, as it selects for only one extreme of the trait, fully functional wings.

Disruptive selection

While stabilising selection makes the average more common, disruptive selection favours the extremes.

The population will become more diverse and may even become separate species. This is called **Speciation.**

This is the least common type of natural selection.





Example

Two distinct male ape types.

Large dominant males

Small stealthy males

Why might this disruptive selection occur?

Example

Two distinct male types.

Large dominant males who are able to copulate and reproduce by force.

Small stealthy males who are able to avoid dominant males and copulate with females.

Middle size males are over powered by dominant males and are not small enough to copulate by stealth.



Two oyster colours dark and light

Why might this disruptive selection happen?

Two oyster colours dark and light

Why might this disruptive selection happen?

Camouflage

Sand, rocks, light, shade



1. A population of deer has legs of very different lengths. A predator enters the area, and can catch the deer with the shorter legs easier than it can catch long-legged deer. Which of the following is true?

A. This population is under directional selection, and the variance is high to begin with.

B. The variance in this population is low. Two species will come from this disruptive selection.

C. The population is under stabilizing selection for long legs.

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A is correct. The pressure on the population is to evolve longer legs. As long as longer legs are equivalent to higher reproduction, legs will get longer. At some point, the benefits of long legs may not outweigh other pressures to reduce leg size. This would be stabilizing selection. The variance in this population is high to begin with, because legs of all sizes are present. With the presence of the predator, the variance will decrease.



2. You are growing beans for the science fair and want to demonstrate disruptive selection. In these beans, you known that a single gene encodes for the color of the bean. R is the red allele, while W is the white allele. These alleles are codominant. Homozygous (RR) individuals produce red beans, while homozygous WW individuals produce white beans. Heterozygous (RW) individuals produce pinkish beans, due to different layers of the bean being different colors. Which of the following experiments would display disruptive selection?

A. Plant the beans in the garden and watch what happens to the frequency of the alleles.

B. Introduce a predator the population, which only eats pink beans.

C. Distribute the beans in a field, and pick only the red and white beans while leaving the pink for the next generation.

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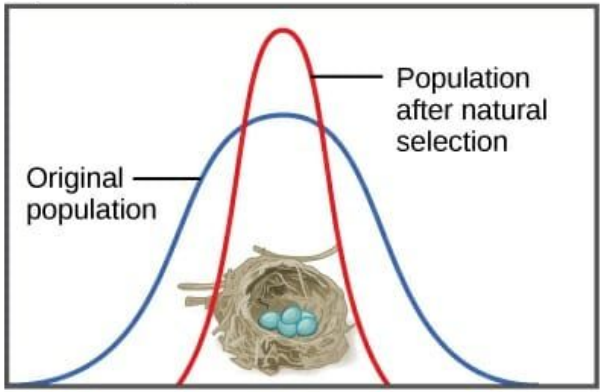
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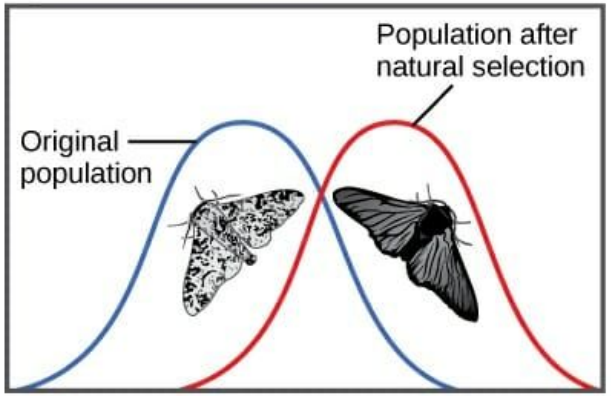
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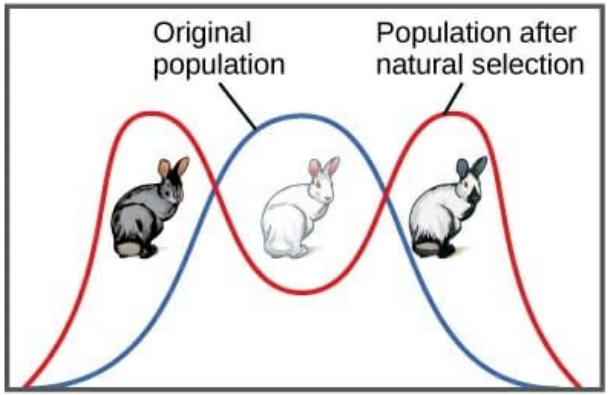
C. Distribute the beans in a field, and pick only the red and white beans while leaving the pink for the next generation.

B is correct. The only experiment that will show disruptive selection is B. If a predator selects against the intermediate trait, the homozygous individuals will increase and form distinct populations over time. In experiment A, nothing is likely to happen as each bean is just as likely to reproduce. In experiment C, the intermediates are being selected for, a case of stabilizing selection.

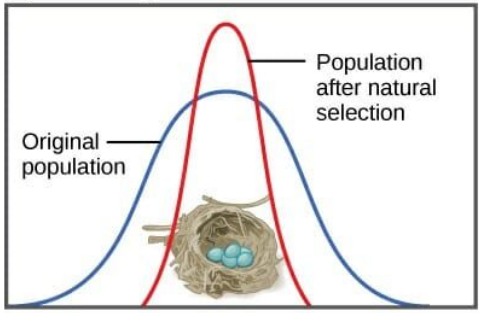
Explain the following three diagrams





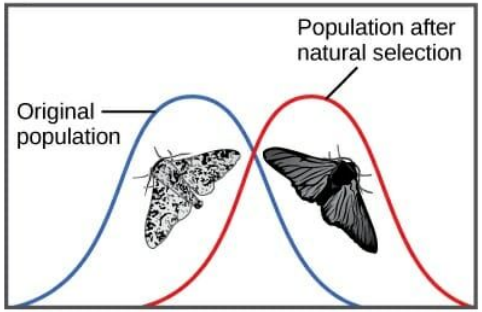


(a) **Stabilizing selection**



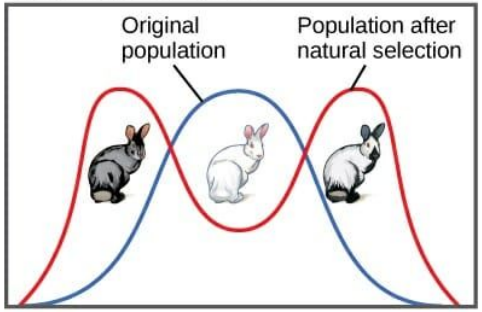
Robins typically lay four eggs, an example of stabilizing selection. Larger clutches may result in malnourished chicks, while smaller clutches may result in no viable offspring.

(b) **Directional selection**



Light-colored peppered moths are better camouflaged against a pristine environment; likewise, dark-colored peppered moths are better camouflaged against a sooty environment. Thus, as the Industrial Revolution progressed in nineteenth-century England, the color of the moth population shifted from light to dark, an example of directional selection.

(c) **Diversifying selection**



In a hyphothetical population, gray and Himalayan (gray and white) rabbits are better able to blend with a rocky environment than white rabbits, resulting in diversifying selection.

D4.1.13—Hardy–Weinberg equation and calculations of allele or genotype frequencies

If a population is stable (not evolving) then the frequency of alleles can be calculated.

In principle you can tell how many of each genotype and phenotype is present in a population based on the Hardy Weinberg Equation

$(p + q)^2 = p^2 + 2pq + q^2 = 1$

Where:

p = the frequency of allele A

q = the frequency of allele a

p^2 = the frequency of individual AA

q^2 = the frequency of individual aa

$2pq$ = the frequency of individual Aa

How to derive Hardy Weinberg equation



If there are two alleles for a particular gene then the sum total of all alleles in the population is the total number of dominant alleles (p) + total number of recessive alleles (q)

If 1 is 100% then $p + q = 1$

We each have two alleles for each gene because two haploid gametes are crossed together. $(p + q) \times (p + q)$

(it may help to think of frequencies as probabilities)

The sum of all diploid cells is also 100%, so $(p + q)^2 = 1$

Can you expand the brackets?

$(p + q)^2 = p^2 + 2pq + q^2$

Frequency of different genotypes in population

p^2 = homozygous dominant

2pq = heterozygous

q^2 = homozygous recessive

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

Example



You have sampled a population in which you know that the percentage of the homozygous recessive genotype (aa) is 36%. Using that 36%, calculate the following:

1. **The frequency of the "aa" genotype.**
2. **The frequency of the "a" allele.**
3. **The frequency of the "A" allele.**
4. **The frequencies of the genotypes "AA" and "Aa."**
5. **The frequencies of the two possible phenotypes if "A" is completely dominant over "a."**

$p + q = 1$

$p^2 + 2pq + q^2 = 1$

$p = A$
$q = a$

$p^2 = AA$
$2pq = Aa$
$q^2 = aa$

$aa = 0.36$
$q^2 = 0.36$
$q = \sqrt{0.36}$
$q = 0.6$
$\therefore A = 60\%$

$p + q = 1$
$p = 1 - 0.6$
$p = 0.4$
$\therefore a = 40\%$

$p^2 = 0.4^2$
$p^2 = 0.16$
$\therefore AA = 16\%$

$q^2 = 0.36$
$\therefore aa = 36\%$

$2pq = 2 \times 0.4 \times 0.6$
$2pq = 0.48$
$\therefore Aa = 48\%$

Classwork



The allele for straight hair is recessive.

Given our class is a stable, non evolving population calculate the likely genotype and phenotype make up of the class for this trait.

Classwork



The allele for the ability to roll one's tongue is dominant.

Given our class is a stable, non evolving population calculate the likely genotype and phenotype make up of the class for this trait.



If calculated an expected number of individuals of each phenotype/ genotype from the Hardy Weinberg equation, how could you see if the actual number observed in a population matched closely enough for it to be said to be in “Hardy Weinberg equilibrium”



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Chi squared test!!!

If there was a significant difference between the observed and the expected, what does this tell you about the population?

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Chi squared test!!!

If there was a significant difference between the observed and the expected, what does this tell you about the population?

That it is not stable, it is evolving (5 methods of evolution)

Extra questions

1. https://www.k-state.edu/parasitology/biology198/hardwein.html

D4.1.14—Hardy–Weinberg conditions that must be maintained for a population to be in genetic equilibrium



Can you describe evolution in terms of the gene pool?

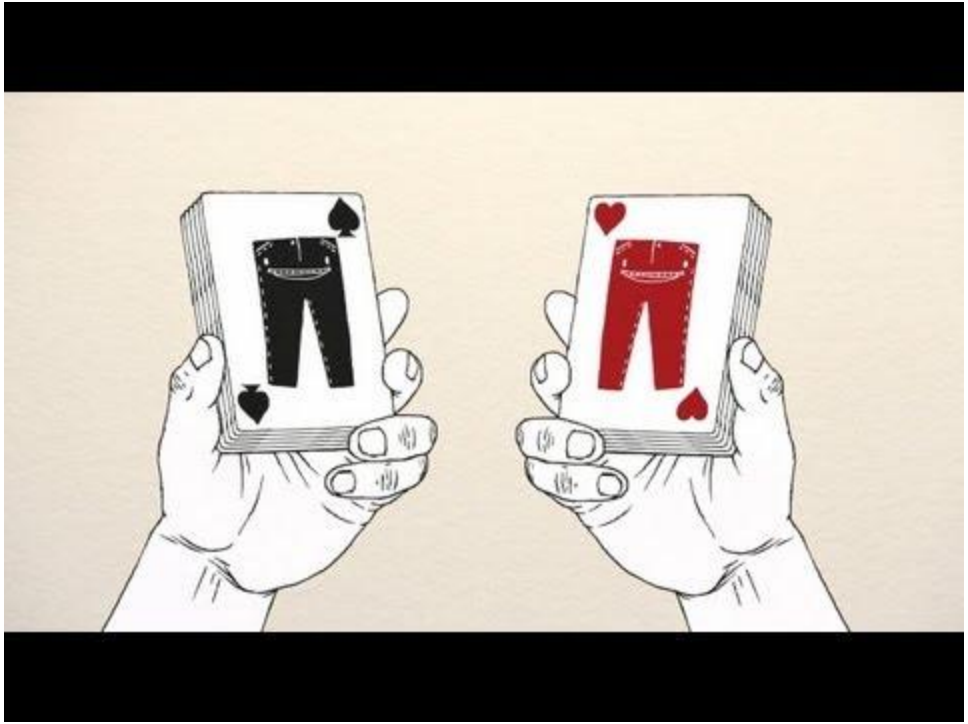
Can you describe evolution in terms of the gene pool?

Evolution is the permanent change of the frequency of an allele within the gene pool.

Natural selection is not the only reasons why the frequency of an allele may change over time.

5 mechanisms of evolution:

1. Genetic drift
2. Non random mating
3. Mutation
4. Gene flow
5. Natural selection



Genetic drift

Not natural selection.

Regardless of whether trait confers benefit

Due to random chance

“Sampling error”

More likely with small populations

Can be seen in **population bottlenecks** and the **founder effect**

Random sampling and genetic drift



Original population

Population bottleneck

An environmental disaster can lead to a dramatically reduced population size.

The few remaining survivors have a reduced amount a variance.

Some alleles will be lost.

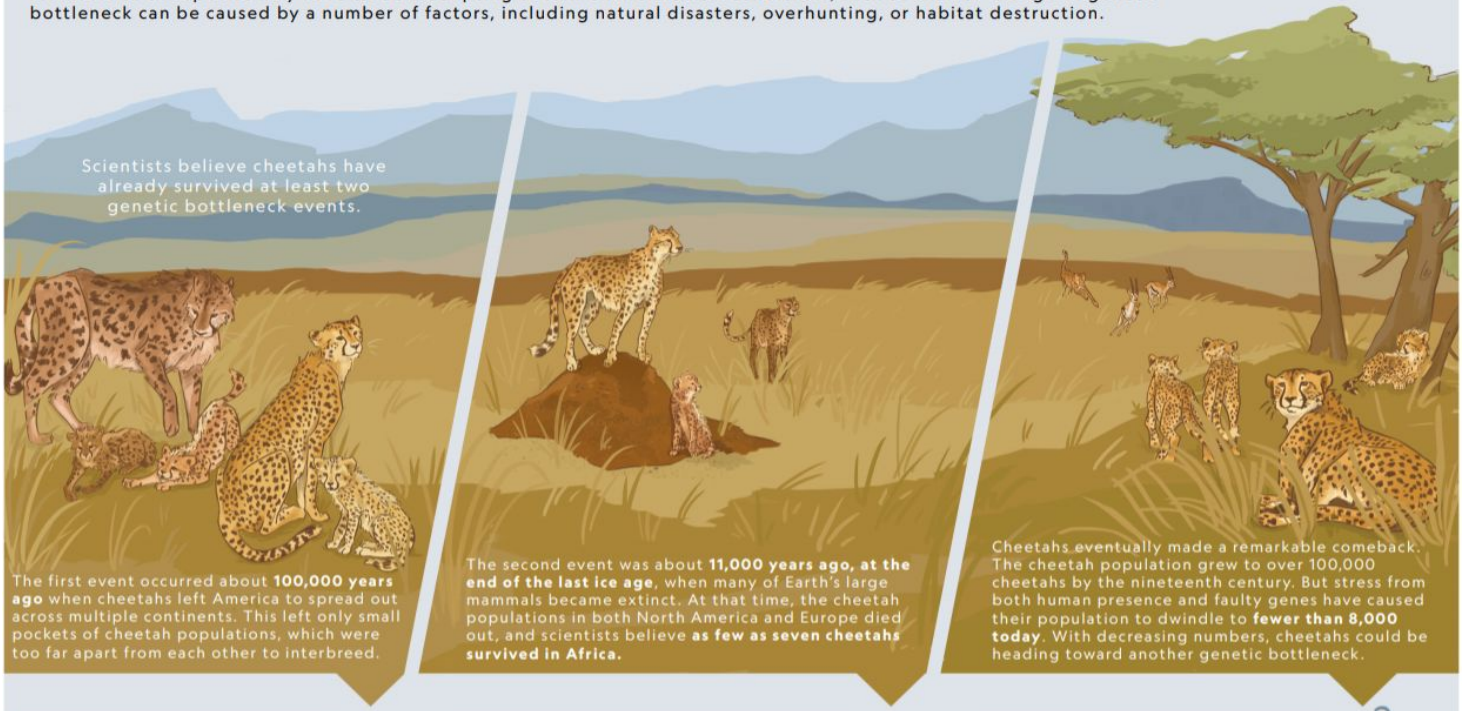
The subsequent generations will therefore also share this limited variance.

GENETIC BOTTLENECK

A Threat to Cheetah Survival

NATIONAL GEOGRAPHIC

A genetic bottleneck occurs when a population is greatly reduced in size. The bottleneck limits the genetic diversity of the species because only a small part of the original population survives. With a greatly reduced gene pool, the remaining members of the species may have trouble adapting to new environmental conditions, such as climate change. A genetic bottleneck can be caused by a number of factors, including natural disasters, overhunting, or habitat destruction.




NATIONAL GEOGRAPHIC
EXPLORERS
IN THE FIELD



In many cultures it is not unusual to be married to one’s cousin. Discuss the implications of this.

Founder effect

Loss of variation when a small group moves to isolated area and start new population.

Forced to reproduce with genetically similar individuals (relatives).

Leads to lack of variation

If one of founders had genetic disease this will be very prevalent in the population

If none of the five causes for evolution are present, the population is said to be in **Hardy Weinberg equilibrium**



Draw around your hand.

Illustrate the five causes of evolution.

D4.1.15—Artificial selection by deliberate choice of traits

Artificial selection

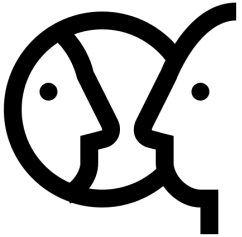
It was known long before Darwin that breeding crops or animals based on desired characteristics could lead to a significant change in their traits.



Bacterial resistance is man made but is not artificial selection.

Bacterial resistance evolves through **natural selection**

Can you...



Compare and contrast population bottleneck and founder effect?

Describe the different patterns of natural selection?

Know when and how to use the Hardy-Weinberg equation?

Describe the different mechanisms of evolution?